

```
yuktha@LAPTOP-UAVR6HQ0:~$ cd OSLab
yuktha@LAPTOP-UAVR6HQ0:~/OSLab$ gedit memory.c

** (gedit:5489): WARNING **: 06:55:58.240: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:5489): WARNING **: 06:55:58.240: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yuktha@LAPTOP-UAVR6HQ0:~/OSLab$ gcc memory.c -o memory
yuktha@LAPTOP-UAVR6HQ0:~/OSLab$ ls
OSLab Practical-5 Practical6 link.c memory.c ossp_q2 ossp_skill2 permission.c shell_history skill_3
Pipes Practical1 TestFolder memory ossp_4 ossp_q3 permission sample.txt signals
yuktha@LAPTOP-UAVR6HQ0:~/OSLab$ ./memory
PID      : 7045
Code address : 0x576d56f0c1c9
Global address : 0x576d56f0f010
Static address : 0x576d56f0f014
Heap address  : 0x576d93dc8010
Stack address : 0x7fff86e33bfc

Process is running...
Press Ctrl+C to terminate.
^C
yuktha@LAPTOP-UAVR6HQ0:~/OSLab$ |
```

```
yuktha@LAPTOP-UAVR6HQ0:~$ cat /proc/7045/maps
576d56f0b000-576d56f0c000 r--p 00000000 08:30 14469 /home/yuktha/OSLab/memory
576d56f0c000-576d56f0d000 r-xp 00001000 08:30 14469 /home/yuktha/OSLab/memory
576d56f0d000-576d56f0e000 r--p 00002000 08:30 14469 /home/yuktha/OSLab/memory
576d56f0e000-576d56f0f000 r--p 00002000 08:30 14469 /home/yuktha/OSLab/memory
576d56f0f000-576d56f10000 rw-p 00003000 08:30 14469 /home/yuktha/OSLab/memory
576d93dc8000-576d93de9000 rw-p 00000000 00:00 0 [heap]
7072d1a00000-7072d1a28000 r--p 00000000 08:30 13641 /usr/lib/x86_64-linux-gnu/libc.so.6
7072d1a28000-7072d1bc0000 r-xp 00028000 08:30 13641 /usr/lib/x86_64-linux-gnu/libc.so.6
7072d1bc0000-7072d1c0e000 r--p 001c0000 08:30 13641 /usr/lib/x86_64-linux-gnu/libc.so.6
7072d1c0e000-7072d1c12000 r--p 0020d000 08:30 13641 /usr/lib/x86_64-linux-gnu/libc.so.6
7072d1c12000-7072d1c14000 rw-p 00211000 08:30 13641 /usr/lib/x86_64-linux-gnu/libc.so.6
7072d1c14000-7072d1c21000 rw-p 00000000 00:00 0
7072d1c5c000-7072d1c5f000 rw-p 00000000 00:00 0
7072d1c66000-7072d1c68000 rw-p 00000000 00:00 0
7072d1c68000-7072d1c6c000 r--p 00000000 00:00 0 [vvar]
7072d1c6c000-7072d1c6e000 r--p 00000000 00:00 0 [vvar_vclock]
7072d1c6e000-7072d1c70000 r-xp 00000000 00:00 0 [vdso]
7072d1c70000-7072d1c71000 r--p 00000000 08:30 13638 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7072d1c71000-7072d1ca0000 r-xp 00001000 08:30 13638 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7072d1ca0000-7072d1cab000 r--p 00030000 08:30 13638 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7072d1cab000-7072d1cad000 r--p 0003b000 08:30 13638 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7072d1cad000-7072d1cae000 rw-p 0003d000 08:30 13638 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7072d1cae000-7072d1caf000 rw-p 00000000 00:00 0
7fff86e15000-7fff86e36000 rw-p 00000000 00:00 0 [stack]
```

yuktha@LAPTOP-UAVR6HQ0:~\$ pmap 7045

7045:	./memory			
0000576d56f0b000	4K	r----	memory	
0000576d56f0c000	4K	r-x--	memory	
0000576d56f0d000	4K	r----	memory	
0000576d56f0e000	4K	r----	memory	
0000576d56f0f000	4K	rw---	memory	
0000576d93dc8000	132K	rw---	[anon]	
00007072d1a00000	160K	r----	libc.so.6	
00007072d1a28000	1632K	r-x--	libc.so.6	
00007072d1bc0000	312K	r----	libc.so.6	
00007072d1c0e000	16K	r----	libc.so.6	
00007072d1c12000	8K	rw---	libc.so.6	
00007072d1c14000	52K	rw---	[anon]	
00007072d1c5c000	12K	rw---	[anon]	
00007072d1c66000	8K	rw---	[anon]	
00007072d1c68000	16K	r----	[anon]	
00007072d1c6c000	8K	r----	[anon]	
00007072d1c6e000	8K	r-x--	[anon]	
00007072d1c70000	4K	r----	ld-linux-x86-64.so.2	
00007072d1c71000	188K	r-x--	ld-linux-x86-64.so.2	
00007072d1ca0000	44K	r----	ld-linux-x86-64.so.2	
00007072d1cab000	8K	r----	ld-linux-x86-64.so.2	
00007072d1cad000	4K	rw---	ld-linux-x86-64.so.2	
00007072d1cae000	4K	rw---	[anon]	
00007fff86e15000	132K	rw---	[stack]	
total	2768K			

yuktha@LAPTOP-UAVR6HQ0:~\$ cat /proc/7045/status

```
Name:      memory
Umask:     0022
State:     S (sleeping)
Tgid:      7045
Ngid:      0
Pid:       7045
PPid:      5443
TracerPid: 0
Uid:       1000    1000    1000    1000
Gid:       1000    1000    1000    1000
FDSize:    256
Groups:    4 24 27 30 46 100 1000
NStgid:    7045
NSpid:     7045
NSpgid:    7045
NSsid:     5443
Kthread:   0
VmPeak:    2768 kB
VmSize:    2768 kB
VmLck:     0 kB
VmPin:     0 kB
VmHWM:     1880 kB
VmRSS:     1880 kB
RssAnon:           96 kB
RssFile:          1784 kB
RssShmem:           0 kB
VmData:    224 kB
VmStk:     132 kB
VmExe:      4 kB
VmLib:    1828 kB
VmPTE:     48 kB
VmSwap:     0 kB
HugetlbPages:      0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xffffffffffffffff
Threads:     1
SigQ:       1/30929
SigPnd:     0000000000000000
ShdPnd:     0000000000000000
SigBlk:     0000000000000000
SigIgn:     0000000000000000
SigCgt:     0000000000000000
CapInh:     0000000000000000
CapPrm:     0000000000000000
CapEff:     0000000000000000
CapBnd:     000001ffffffffffff
```

```
CapAmb: 0000000000000000  
NoNewPrivs: 0  
Seccomp: 0  
Seccomp_filters: 0  
Speculation_Store_Bypass: thread vulnerable  
SpeculationIndirectBranch: conditional enabled  
Cpus_allowed: 3ffff  
Cpus_allowed_list: 0-17  
Mems_allowed: 00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,  
000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000001  
Mems_allowed_list: 0  
voluntary_ctxt_switches: 176  
nonvoluntary_ctxt_switches: 0  
yuktha@LAPTOP-UAVR6HQO:~$
```